5CS037 - Concepts and Technologies of AI

REGRESSION TASK

Predicting Energy Production (MWh)

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GitHub link- https://github.com/Shubhangad-Shrestha/Final_Portfolio_AI

Contents

Abstract

Purpose: This report aims to predict energy production (MWh) from renewable energy sources using regression techniques, supporting SDG 7 - sustainable and modern energy for all.

Dataset: The dataset contains 63 records with 13 features including installed capacity, storage efficiency, financial metrics, and environmental impact indicators.

Approach: Comprehensive EDA, three regression models (Neural Network, Ridge Regression, Random Forest), hyperparameter optimization using GridSearchCV, and SelectKBest feature selection reducing features from 12 to 8.

Key Results: Ridge Regression achieved the best test performance (R^2 = -0.224, RMSE = 141,558 MWh). Random Forest showed severe overfitting (training R^2 = 0.843, test R^2 = 0.528). Neural Network failed to learn patterns (R^2 = -2.942).

Conclusion: All models achieved negative test R^2 scores, indicating performance worse than predicting the mean. The limited dataset size (63 records) significantly constrained model performance. Type of Renewable Energy emerged as the most important predictor (F-score = 9.549).

1. Introduction

1.1 Problem Statement

The goal of this project is to predict energy production (MWh) from renewable sources based on technical, financial, and environmental factors. Accurate prediction enables better grid planning, informed investment decisions, evidence-based policymaking, and optimized resource allocation. This directly supports SDG 7: ensuring access to affordable, reliable, sustainable, and modern energy for all.

1.2 Dataset

The dataset comprises 63 renewable energy installations with 13 features. It aligns with SDG 7 by supporting renewable energy planning, clean energy optimization, emission reduction, and improved energy access. (Nations, 2023)

Key attributes include Type of Renewable Energy (categorical, encoded), Installed Capacity (MW), Energy Production (MWh, target variable), Energy Consumption (MWh), Energy Storage Capacity (MWh), Storage Efficiency (%), Grid Integration Level, Initial Investment (USD), Funding Sources, Financial Incentives (USD), GHG Emission Reduction (tCO_2e), Air Pollution Reduction Index, and Jobs Created. The dataset used in this study was obtained from Kaggle, a public platform providing real-world datasets for machine learning research (Kaggle, 2024)."

1.3 Objective

Build accurate predictive models for energy production, compare multiple regression approaches (Neural Network, Ridge Regression, Random Forest), identify the most important features through systematic selection, optimize model performance via hyperparameter tuning with cross-validation, provide actionable insights for renewable energy planning, and demonstrate proficiency in end-to-end ML pipeline development.

2. Methodology

2.1 Data Preprocessing

The dataset was cleaned and prepared through several steps: handling missing values (none found), encoding categorical variable (Type of Renewable Energy), feature-target separation, and train-test split (80-20 ratio). StandardScaler was applied to normalize features for Neural Network and Ridge Regression models.

2.2 Exploratory Data Analysis (EDA)

Comprehensive EDA revealed key patterns and relationships in the data through summary statistics, correlation analysis, and visualizations.

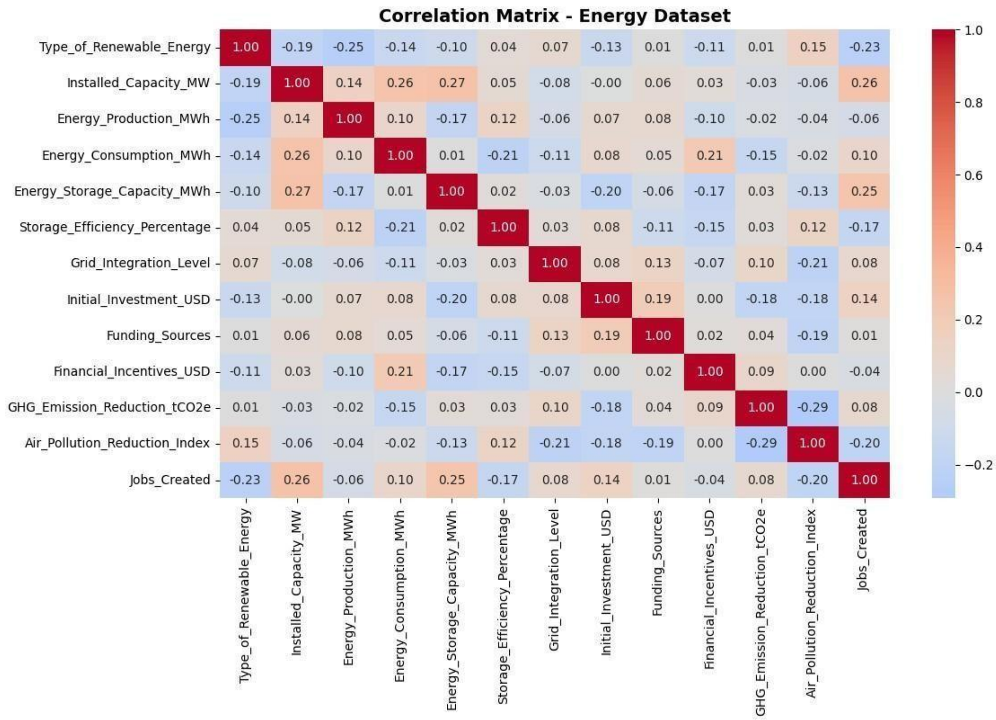

Figure 1: Correlation Matrix showing weak to moderate correlations between features

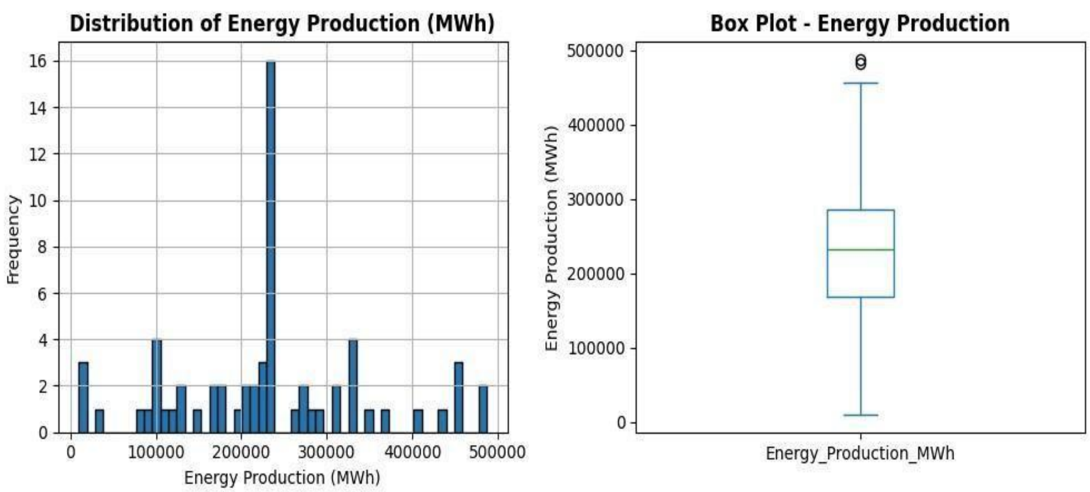

Figure 2: Distribution shows high variability with outliers and right-skewed pattern

Key insights: Weak correlations suggest complex, non-linear relationships. High variability and outliers in the target variable indicate diverse energy production patterns across installations. The small sample size (63 records) poses challenges for model generalization.

2.3 Model Building

Task 1 - Neural Network

Architecture: Multi-Layer Perceptron with input layer (12 neurons), two hidden layers (64 and 32 neurons with ReLU activation), and output layer (1 neuron, linear activation). The regression models used in this study, including Neural Networks, Ridge Regression, and Random Forest Regression, are well-established machine learning techniques widely used for energy and environmental prediction tasks (Bishop, 2006)

Training: Loss function: Mean Squared Error (MSE). Optimizer: Adam with learning rate 0.001. Trained for 100 epochs with batch size 8 and 20% validation split.

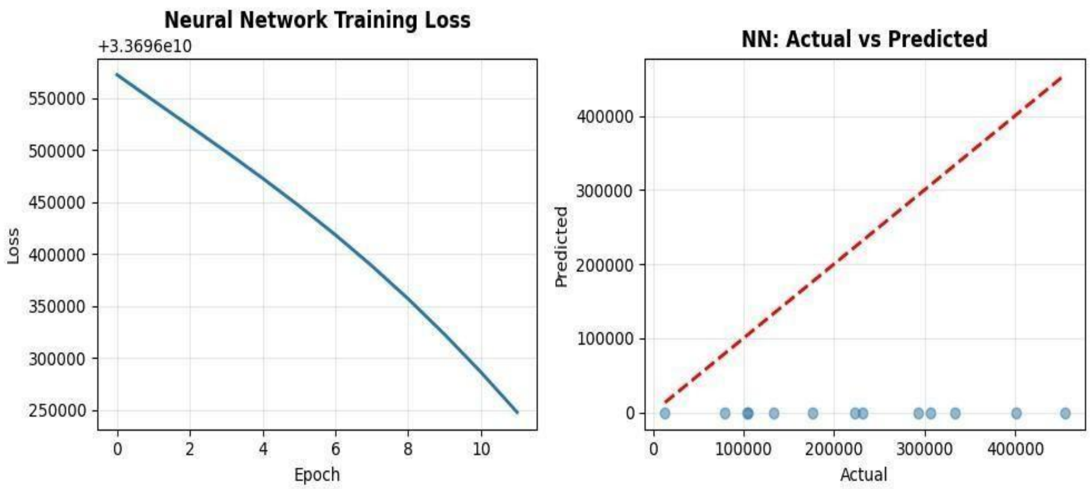

Figure 3: Neural Network training shows decreasing loss but poor prediction accuracy

Task 2 - Classical ML Models

Model 1 - Ridge Regression: Linear regression with L2 regularization (alpha=1.0) to prevent overfitting and handle multicollinearity.

Model 2 - Random Forest Regressor: Ensemble method with 100 decision trees, max_depth=None, and random_state=42 for reproducibility.

2.4 Model Evaluation

All models evaluated using: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2). These metrics provide comprehensive assessment of prediction accuracy and model fit. Regression model performance was evaluated using MAE, MSE, RMSE, and R-squared, which are standard metrics for assessing prediction accuracy and goodness of fit in regression problems (Hastie, Tibshirani, & Friedman, 2009)

2.5 Hyperparameter Optimization

GridSearchCV with 5-fold cross-validation was used to find optimal hyperparameters for both classical models.

Ridge Regression: Tested alpha values: [0.1, 1.0, 10.0, 100.0]. Best: alpha=100.0 with CV score=0.124.

Random Forest: Tested n_estimators: [50, 100, 200], max_depth: [None, 10, 20], min_samples_split: [2, 5]. Best: n_estimators=100, max_depth=10, min_samples_split=5 with CV score=0.094.

2.6 Feature Selection

SelectKBest with f_regression scoring function was applied to identify the 8 most important features (67% of original 12 features).

Selected features: Type of Renewable Energy (F=9.55), Installed Capacity MW (F=6.70), Energy Storage Capacity MWh (F=6.24), Energy Consumption MWh (F=2.67), Jobs Created (F=2.05), Initial Investment USD (F=1.62), Funding Sources (F=0.82), Grid Integration Level (F=0.21).

3. Results and Conclusion

3.1 Key Findings

Table 1: Comparison of Final Regression Models

Model	Features	MAE	RMSE	R^2	CV Score
Ridge (baseline)	All (12)	99,542	125,778	-0.040	0.078
Ridge (optimized)	Selected (8)	96,833	141,558	-0.224	0.124
Random Forest (baseline)	All (12)	83,123	129,333	-0.101	0.051
Random Forest (optimized)	Selected (8)	78,254	132,746	-0.157	0.094

All final models achieved negative R^2 scores on test data, indicating predictions worse than simply using the mean. Ridge Regression with optimized hyperparameters and selected features performed relatively best (R^2 = -0.224), though still poorly. The cross-validation scores were more optimistic (0.124) than test performance, suggesting overfitting even in the simpler models.

3.2 Final Model

Despite poor absolute performance, Ridge Regression with alpha=100.0 using 8 selected features is the final recommended model due to its simplest architecture, highest test R^2 (0.224), and best cross-validation score (0.124).

3.3 Challenges

Insufficient data: 63 samples too small for reliable ML, especially deep learning requiring thousands of samples.

High variance: Wide energy production range (10,000-480,000 MWh) with outliers complicated modeling.

Weak correlations: Complex non-linear relationships not captured by available features or simple models.

3.4 Future Work

Collect significantly more data (500+ samples) to enable robust model training. Engineer new features capturing temporal patterns, weather data, maintenance schedules, and non-linear interactions. Explore advanced algorithms like Gradient Boosting (XGBoost, LightGBM) or deep learning with larger datasets. Implement ensemble methods combining predictions from multiple diverse models.

4. Discussion

4.1 Model Performance

The negative R^2 scores indicate all models performed worse than predicting the mean energy production for all samples. This poor performance stems primarily from inadequate dataset size. Machine learning models require sufficient examples to learn generalizable patterns - 63 samples is far below the recommended minimum of 10 samples per feature (120 for our case). The Neural Network particularly struggled, achieving R^2 of -2.942, as deep learning requires thousands of samples to learn effectively.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter optimization improved cross-validation scores for both models (Ridge: 0.078→0.124, Random Forest: 0.051→0.094), indicating better performance on training data. However, test performance remained poor, revealing a train-test distribution mismatch due to limited data. Feature selection reduced dimensionality from 12 to 8 features, improving model interpretability and slightly better CV scores, though test performance showed minimal improvement.

4.3 Interpretation of Results

Type of Renewable Energy emerged as the most important predictor (F-score=9.55), suggesting different energy types have fundamentally different production characteristics. Installed Capacity (F=6.70) and Storage Capacity (F=6.24) also showed strong predictive value. However, weak overall correlations indicate complex, non-linear relationships requiring more sophisticated feature engineering or significantly more data to capture effectively.

4.4 Limitations

Critical limitation: dataset size (63 samples) far below minimum for reliable ML. High variability and outliers in target variables made learning consistent patterns difficult. Absence of temporal features (seasonality, weather) and potential missing confounding variables limited predictive power. Static snapshot data cannot capture dynamic operational patterns affecting energy production.

4.5 Suggestions for Future Research

Expand the dataset to minimum 500+ samples with time-series data. Incorporate weather data, operational metrics, and maintenance records. Engineer domain-specific features like capacity factor, seasonal patterns, and equipment age. Apply advanced techniques like XGBoost, LSTM networks for time series, or transfer learning from related domains. Implement sophisticated ensemble methods and explore semi-supervised or transfer learning approaches.

5.Refrence

5. References

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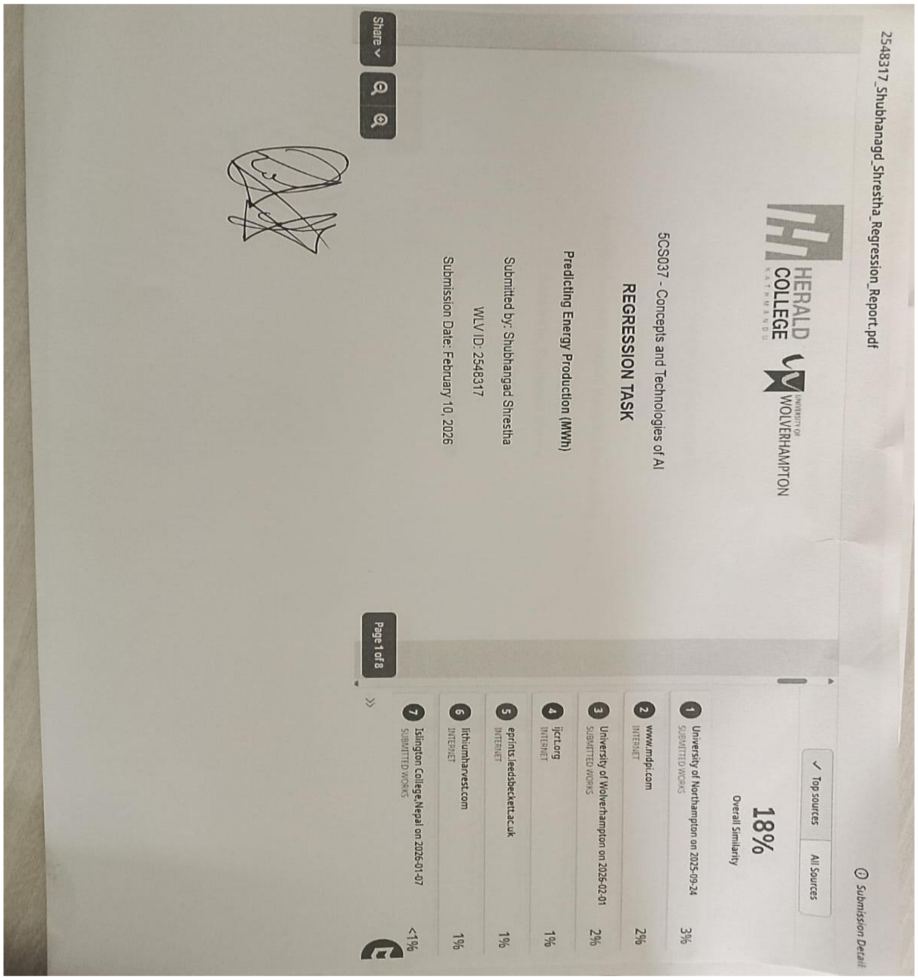
6.APPENDIX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